

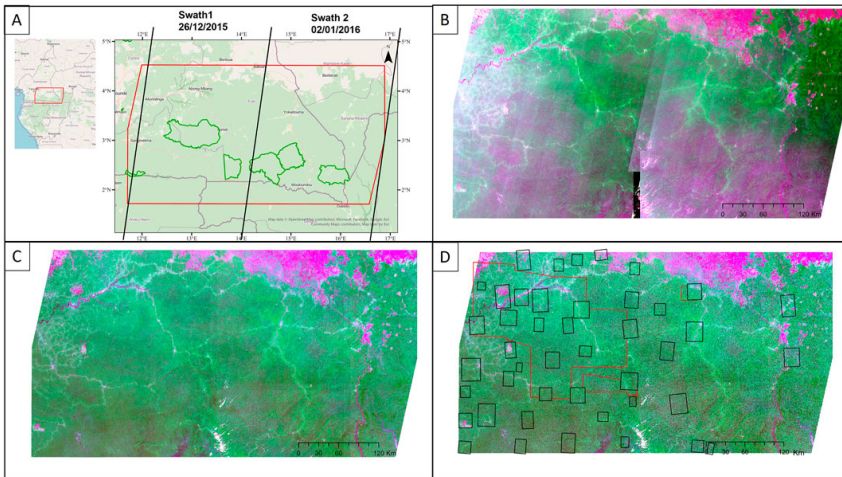
Parameter efficient handling of acquisition artefacts to map tropical forests

Biomonweek 2026

Paul Tresson et al.

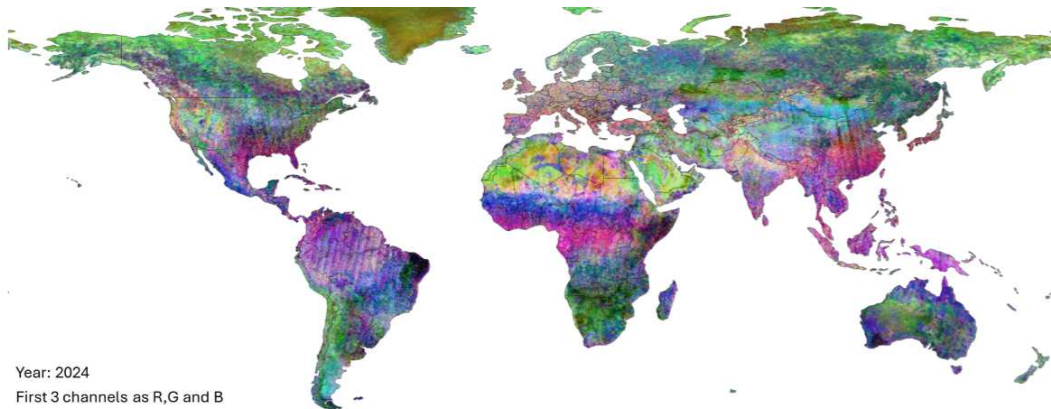
Institut de Recherche pour le Développement - UMR AMAP

Context: Mapping tropical forests is hard



Viennois et al., 2025

Context: Omnipotent foundation models ?



From Feng et al., 2025

Context: Omnipotent foundation models ?

- Not always at local scale
- Not always on niche tasks
- Not for all modalities
- Not always open source and open weights

Can we handle acquisition artefacts to map tropical forests ?

...

Can we handle acquisition artefacts to map tropical forests ?

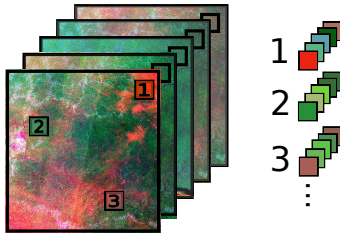
Without training a new expensive model

Constraints

- Robust
- Versatile
- Actionnable without HPC
- Actionnable with limited labelled dataset

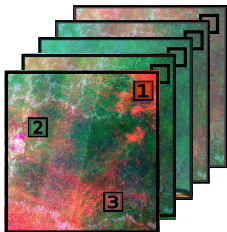
Methods : overview

Sampling

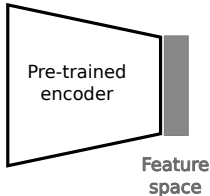


Methods : overview

Sampling



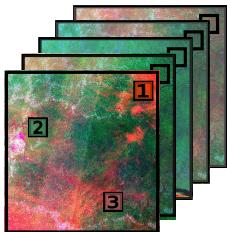
Model



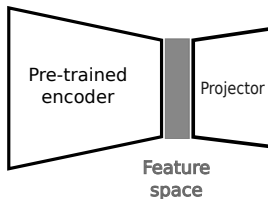
Encoders : DINOv3 and DOFA Siméoni et al., 2025; Xiong et al., 2024

Methods : overview

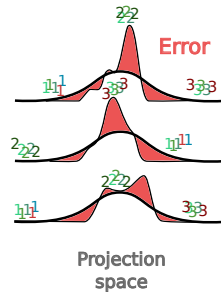
Sampling



Model



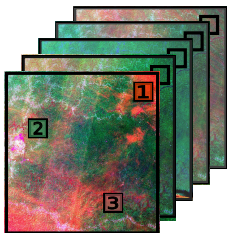
Task



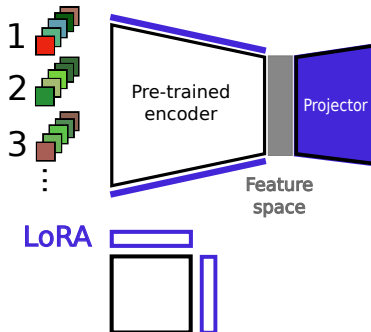
Based on LeJEPA Balestrierio and LeCun, 2025

Methods : overview

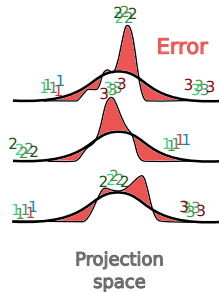
Sampling



Model



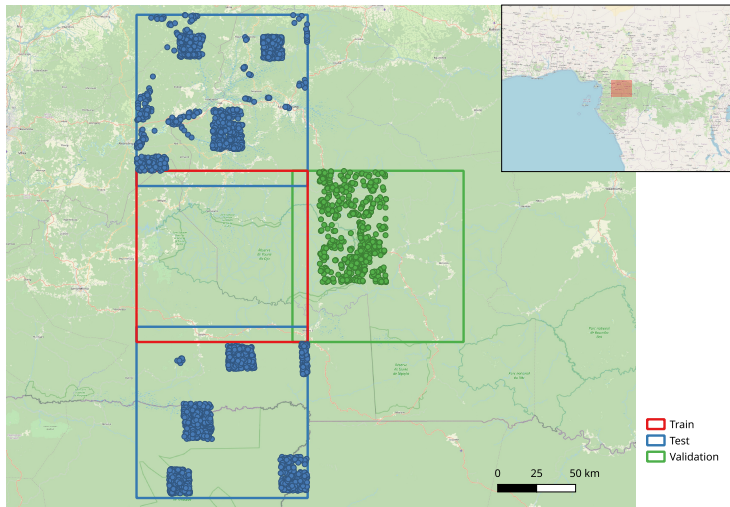
Task



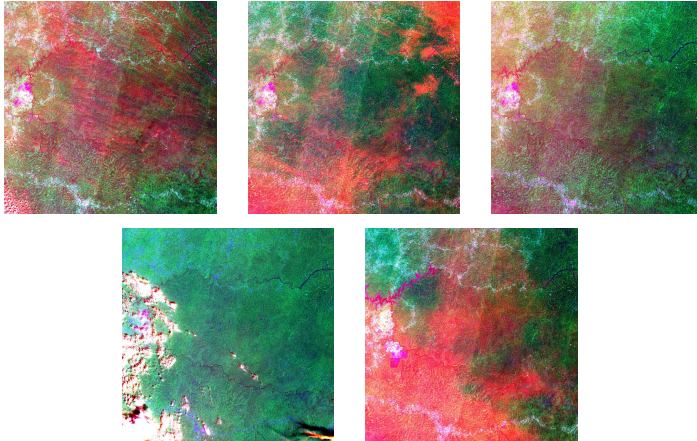
LoRA see Hu et al., 2022

Methods : datasets

- S2 tiles
- 655 labeled validation points
- 2103 labeled test points
- less than 15% clouds



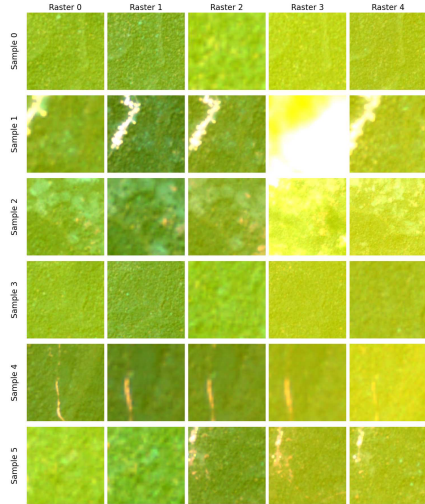
Methods : train dataset



RGB composition with bands 2, 8 and 11

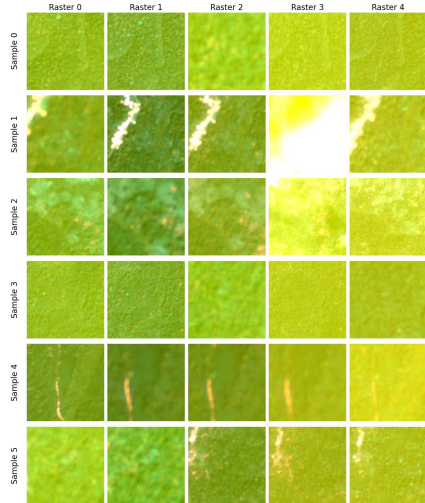
Methods : train sampling

- Random sampling on 5 dates



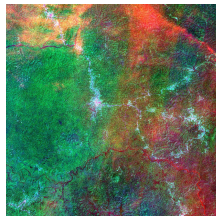
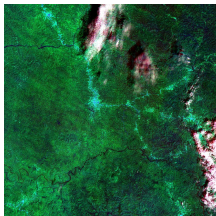
Methods : train sampling

- Random sampling on 5 dates
- Random zoom and crop
(Between 30-150 pixels)

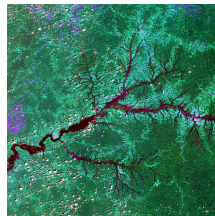
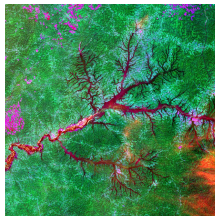
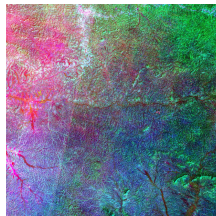
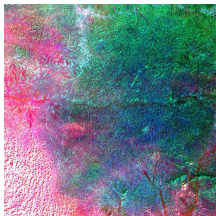


Methods : validation and test datasets

Validation :

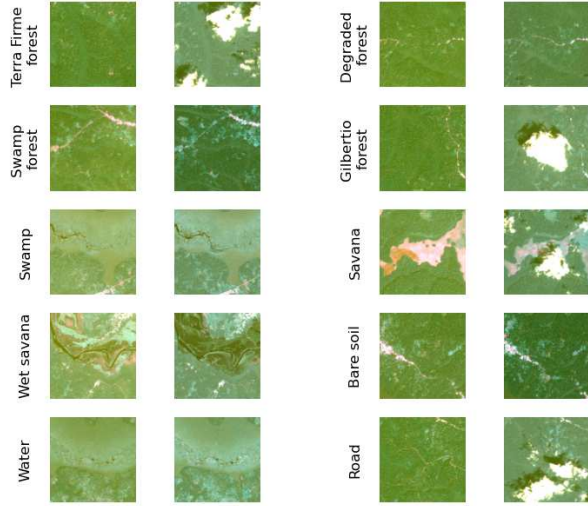


Test :



Methods : validation scheme

- 150 px centered on labeled point



Methods : validation scheme

- 150 px centered on labeled point
- Geographical stratified 5-fold

Terra Firme
forest



Swamp
forest



Swamp



Wet savana



Water



Degraded
forest



Gilbertio
forest



Savana



Bare soil

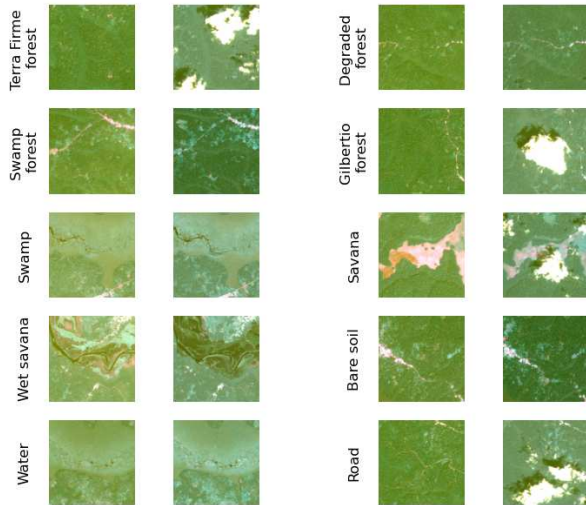


Road



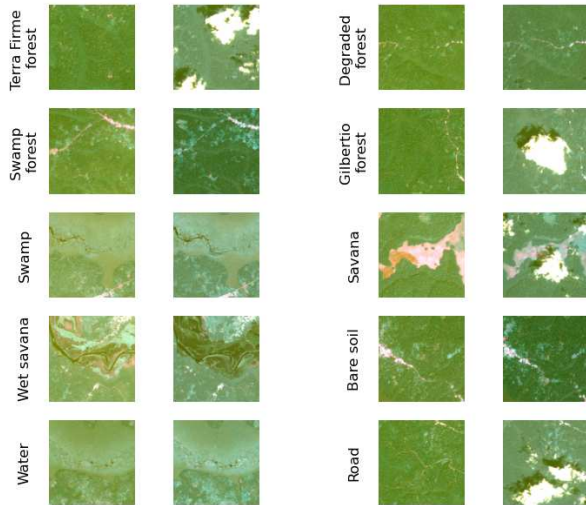
Methods : validation scheme

- 150 px centered on labeled point
- Geographical stratified 5-fold
- KNN fitted on model embedding of version A ...

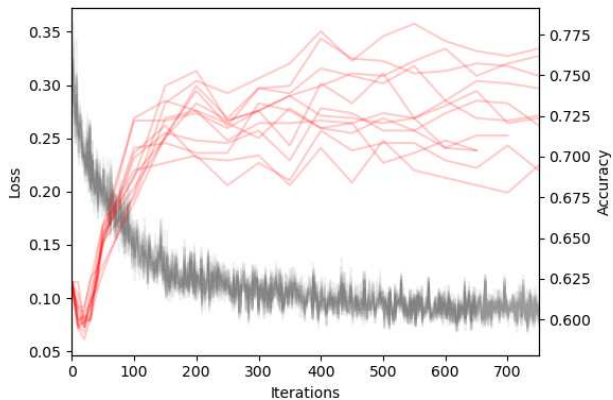


Methods : validation scheme

- 150 px centered on labeled point
- Geographical stratified 5-fold
- KNN fitted on model embedding of version A ...
... and tested on version B



Results : training behaviour



Results : Confusion matrix

Before training

True label \ Predicted label	Build	Degraded Forest	Gilbertio. Forest	Road	Savana	Swamp	Swamp forest	Terra Firme Forest	Water	Wet savana
Build	24	77	1	4	2	1	26	2	0	0
Degraded Forest	28	112	37	2	3	0	99	43	2	0
Gilbertio. Forest	2	7	84	0	0	0	150	43	0	0
Road	8	17	3	1	4	0	9	2	0	0
Savana	2	9	0	2	79	1	6	5	2	8
Swamp	2	2	0	0	7	2	0	0	5	11
Swamp forest	34	160	193	3	12	0	141	155	7	0
Terra Firme Forest	9	36	91	1	3	0	169	105	1	0
Water	0	5	0	0	7	1	9	0	3	2
Wet savana	0	0	0	0	5	6	0	0	1	8

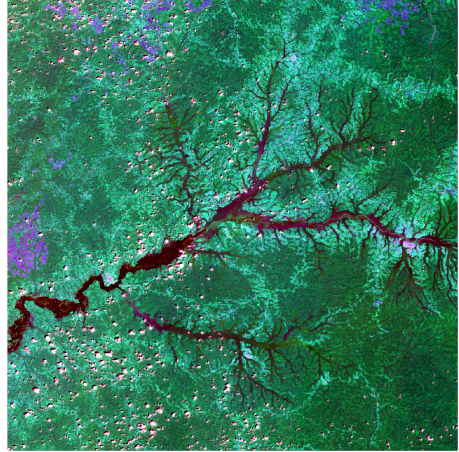
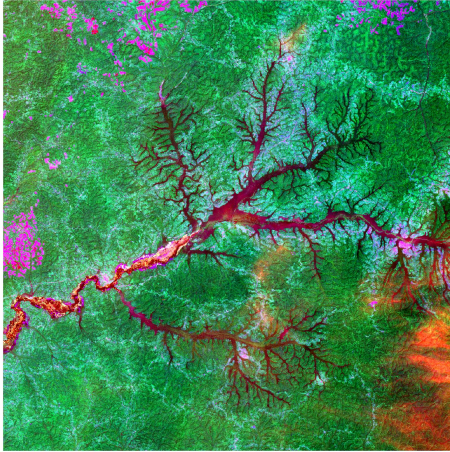
After training

True label \ Predicted label	Build	Degraded Forest	Gilbertio. Forest	Road	Savana	Swamp	Swamp forest	Terra Firme Forest	Water	Wet savana
Build	79	41	0	8	2	0	6	1	0	0
Degraded Forest	39	152	10	4	1	0	52	68	0	0
Gilbertio. Forest	0	3	133	0	0	0	105	45	0	0
Road	26	10	0	2	0	0	2	4	0	0
Savana	1	1	0	3	93	1	1	2	0	12
Swamp	3	0	0	0	9	8	0	0	2	7
Swamp forest	3	22	102	1	0	0	521	38	18	0
Terra Firme Forest	2	24	41	0	0	0	36	312	0	0
Water	0	0	0	0	0	5	4	0	13	5
Wet savana	0	0	0	0	8	4	0	0	2	6

36% accuracy gain !

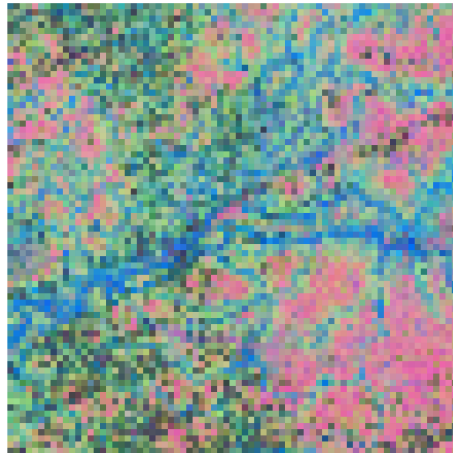
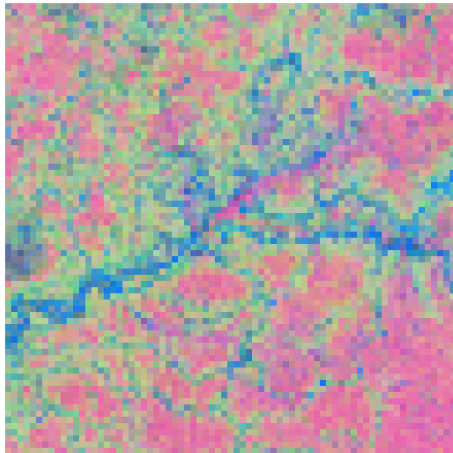
Visualisation

Original rasters



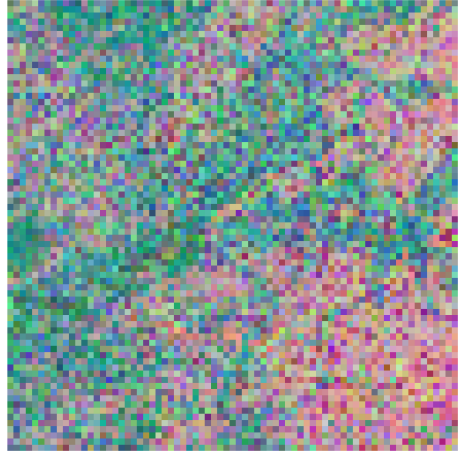
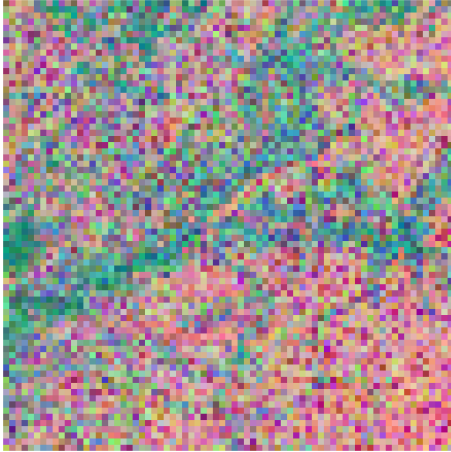
Visualisation : PCA

Before training



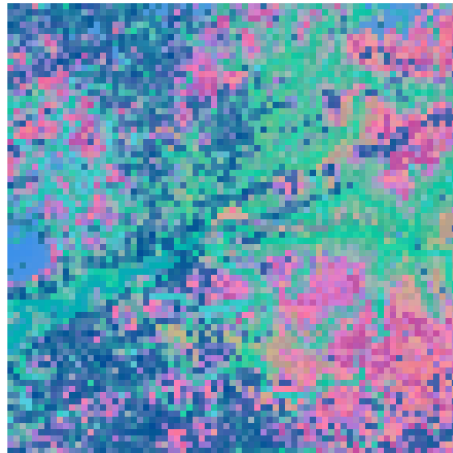
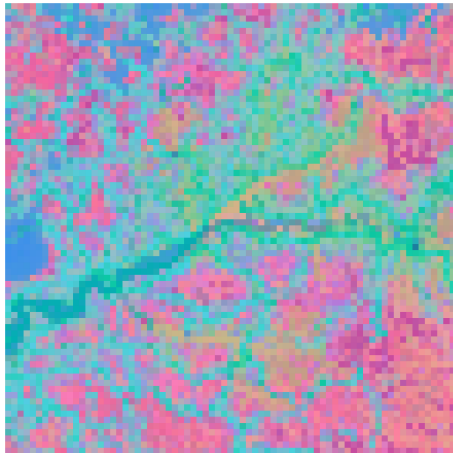
Visualisation : PCA

After training



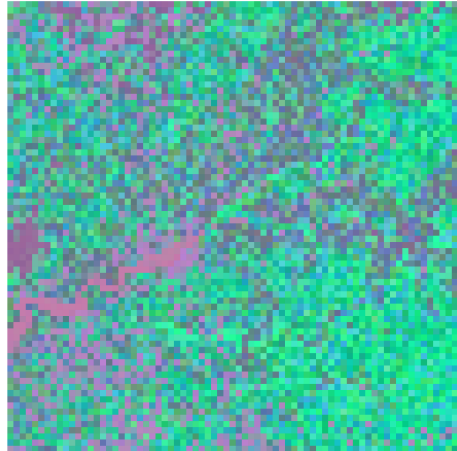
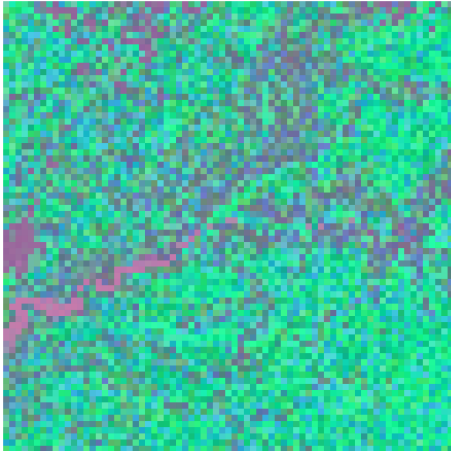
Visualisation : UMAP

Before training



Visualisation : UMAP

After training



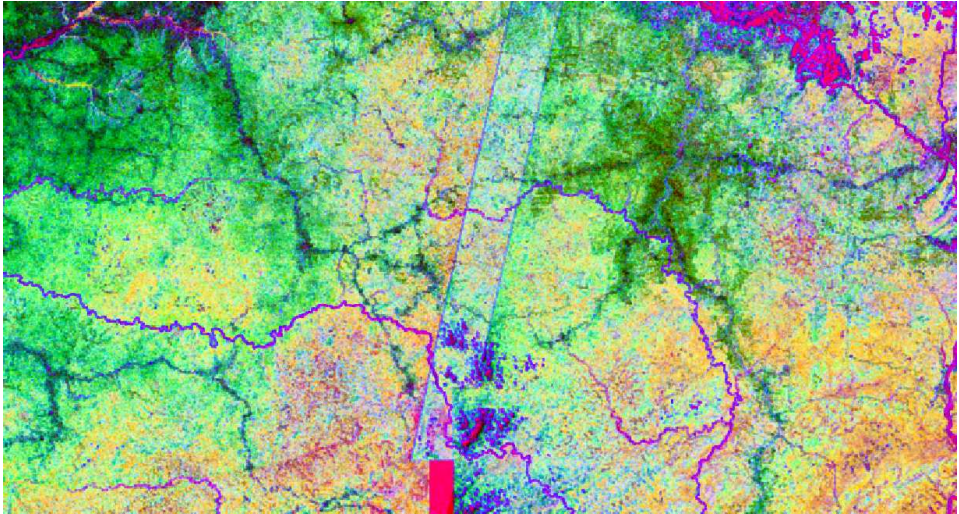
Scaling up

Original



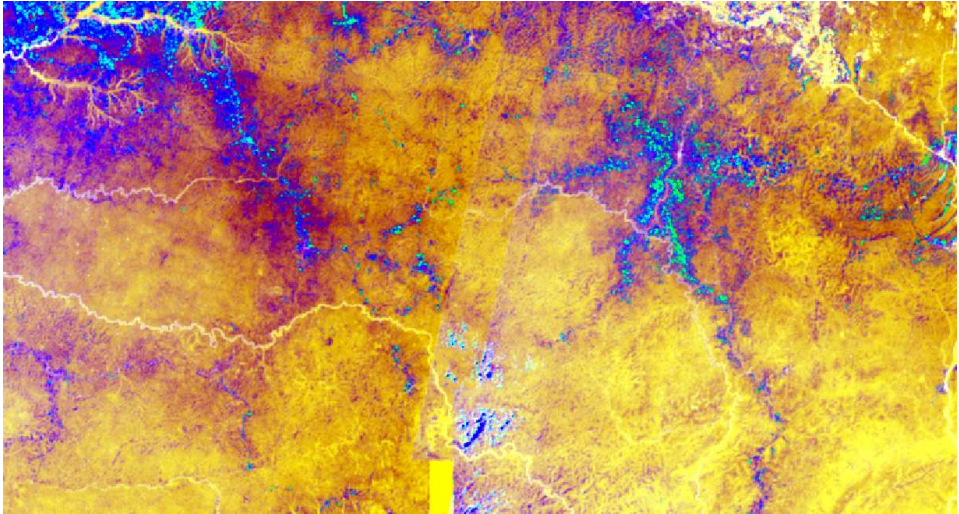
Scaling up

Before training



Scaling up

After training



Discussion

Not perfect yet but

- Only a couple hundred iterations (not epochs)

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0.9 / 85.6M

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- Only a couple hundred iterations (not epochs)
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- Valid for any raster data

Discussion

Not perfect yet but

- Only a couple hundred iterations (not epochs)
- No labels needed
- Training only $\approx 1\%$ of the weights on a single GPU
0.9 / 85.6M
- Valid for any raster data
- To map something **that does not vary between versions**

Thanks for your
attention !

Any questions ?

Download this talk:

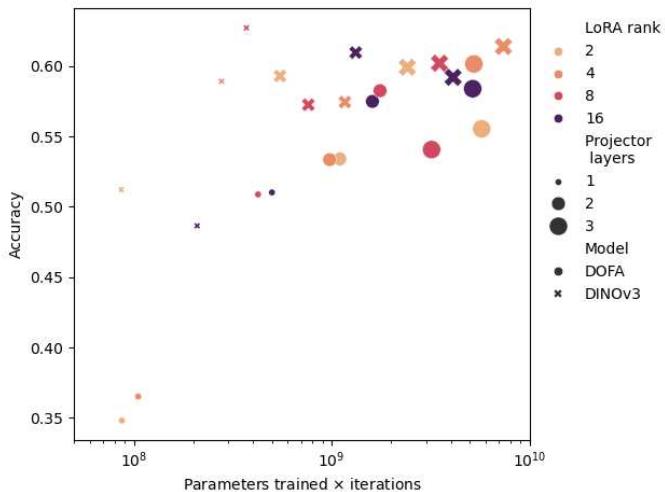


References i

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- Feng, Zhengpeng et al. (2025). **“Tessera: Temporal embeddings of surface spectra for earth representation and analysis”**. In: *arXiv preprint arXiv:2506.20380*.
- Hu, Edward J et al. (2022). **“Lora: Low-rank adaptation of large language models.”**. In: *ICLR 1.2*, p. 3.
- Siméoni, Oriane et al. (2025). **“Dinov3”**. In: *arXiv preprint arXiv:2508.10104*.
- Viennois, Gaëlle et al. (2025). **“Sentinel-2 forest typology mapping in Central Africa: assessing deep learning and image preprocessing effects”**. In: *Frontiers in Remote Sensing 6*, p. 1682132.

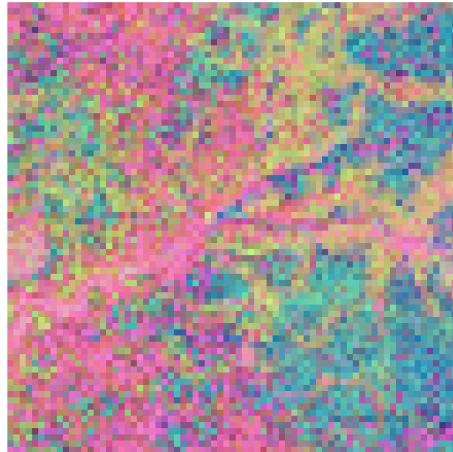
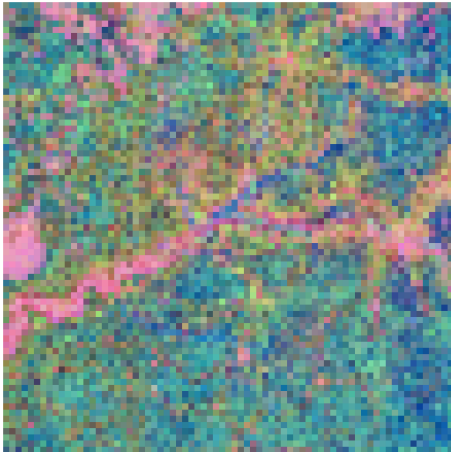
Xiong, Zhitong et al. (2024). “**Neural Plasticity-Inspired Foundation Model for Observing the Earth Crossing Modalities**”. In: *arXiv preprint arXiv:2403.15356*.

Results : Accuracy vs trained parameters



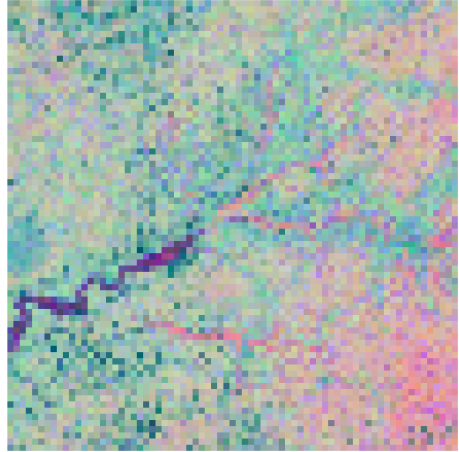
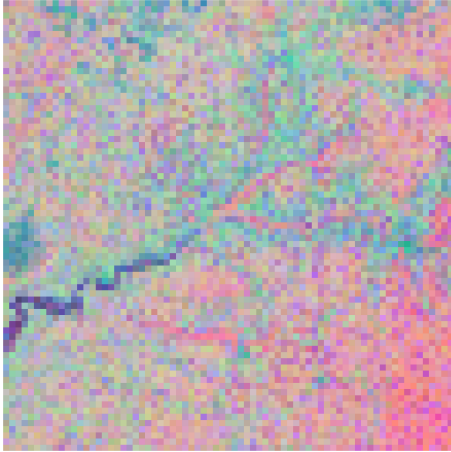
Visualisation DOFA PCA

Before training



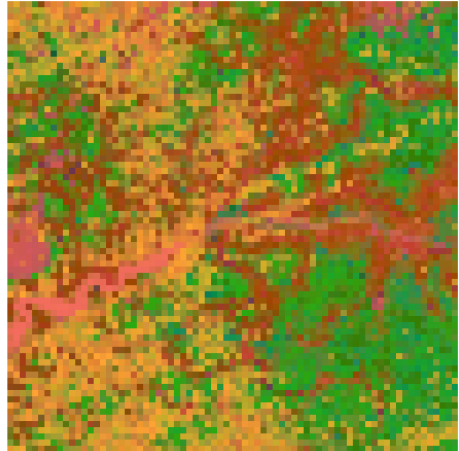
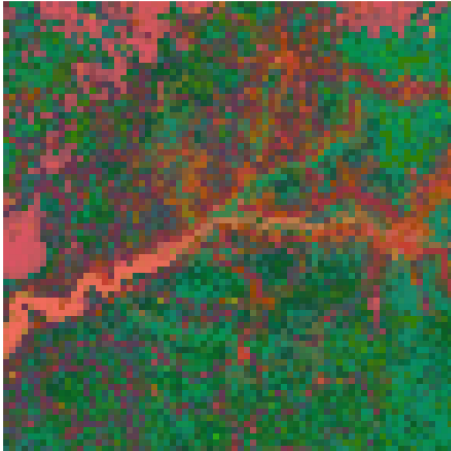
Visualisation DOFA PCA

After training



Visualisation DOFA UMAP

Before training



Visualisation DOFA UMAP

After training

